CYBERSECURITY ACADEMY

Master Notebook • Beginner to Job-Readiness • 12-Month Roadmap

How to use this notebook

Study one lesson at a time. Read the explanation, draw a simple diagram, complete the safe practice task, and answer revision questions without looking. Use the website's Hindi narration when you need to hear the concept aloud.

Study block	What to do
30 minutes	Learn one small concept
45 minutes	Practise on your laptop using your own files or authorised labs
25 minutes	Write definitions, examples, and comparison tables
20 minutes	Recall from memory and answer quiz questions

Ethical rule: practise only on your own devices, your own sample files, or systems/labs where you have explicit permission.

12-Month Roadmap at a Glance

Month	Main focus	Evidence of learning
Month 1	Computer Fundamentals & Digital Safety	Complete lessons, practice tasks, and revision
Month 2	Networking Foundations	Complete lessons, practice tasks, and revision
Month 3	Linux, Windows & Command Line	Complete lessons, practice tasks, and revision
Month 4	Python Foundations	Complete lessons, practice tasks, and revision
Month 5	Security Fundamentals	Complete lessons, practice tasks, and revision
Month 6	Defensive Security & Logs	Complete lessons, practice tasks, and revision
Month 7	Web & Application Security	Complete lessons, practice tasks, and revision
Month 8	Cloud Fundamentals	Complete lessons, practice tasks, and revision
Month 9	Network Analysis & System Hardening	Complete lessons, practice tasks, and revision
Month 10	Containers, DevSecOps & Automation	Complete lessons, practice tasks, and revision
Month 11	Portfolio & Capstone	Complete lessons, practice tasks, and revision
Month 12	Job Preparation & Review	Complete lessons, practice tasks, and revision

Month 1 — Computer Fundamentals & Digital Safety

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Computer basics

Simple explanation: A computer receives input, processes data using instructions, stores data, and produces output. Main stages: Input → Processing → Storage → Output.

Study part	Your notebook answer
Definition	Computer basics means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Hardware and software

Simple explanation: Hardware is the physical part (keyboard, screen, RAM, SSD). Software is the instructions/programs (Windows, browser, Python).

Study part	Your notebook answer
Definition	Hardware and software means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Files and operating systems

Simple explanation: An operating system manages hardware, files, users, applications, and permissions. Learn folders, file extensions, updates, and backups.

Study part	Your notebook answer
Definition	Files and operating systems means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Accounts and safe habits

Simple explanation: Use unique passwords, multi-factor authentication (MFA), screen lock, updates, backups, and cautious link handling.

Study part	Your notebook answer
Definition	Accounts and safe habits means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 2 — Networking Foundations

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Network basics

Simple explanation: A network connects devices so they can exchange data. LAN covers a local area; WAN covers a wide area. The Internet is a global network of networks.

Study part	Your notebook answer
Definition	Network basics means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: IP and MAC addresses

Simple explanation: An IP address helps route data between networks. A MAC address identifies a network interface on a local network. They serve different purposes.

Study part	Your notebook answer
Definition	IP and MAC addresses means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: DNS, DHCP, ports

Simple explanation: DNS maps domain names to IP addresses. DHCP can assign network settings automatically. Ports help a device direct traffic to the right service.

Study part	Your notebook answer
Definition	DNS, DHCP, ports means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: TCP, UDP, HTTP(S)

Simple explanation: TCP provides ordered, reliable delivery; UDP has lower overhead but no built-in delivery guarantee. HTTPS protects web traffic in transit using TLS.

Study part	Your notebook answer
Definition	TCP, UDP, HTTP(S) means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 3 — Linux, Windows & Command Line

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Linux basics

Simple explanation: Linux is an operating-system family used on servers and security tools. Practise only in your own system or authorised lab.

Study part	Your notebook answer
Definition	Linux basics means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Files and permissions

Simple explanation: Learn paths, directories, read/write/execute permissions, users, groups, and least privilege.

Study part	Your notebook answer
Definition	Files and permissions means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Windows security

Simple explanation: Understand updates, Defender, firewall, users, event logs, and basic system settings.

Study part	Your notebook answer
Definition	Windows security means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Command-line habits

Simple explanation: Use commands carefully. Understand a command before running it; avoid commands copied from unknown sources.

Study part	Your notebook answer
Definition	Command-line habits means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 4 — Python Foundations

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Core syntax

Simple explanation: Learn variables, strings, numbers, booleans, input/output, operators, and type conversion.

Study part	Your notebook answer
Definition	Core syntax means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Decisions and loops

Simple explanation: if/elif/else selects a path. for and while repeat work. Use clear indentation and test boundary cases.

Study part	Your notebook answer
Definition	Decisions and loops means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Collections and functions

Simple explanation: Lists hold ordered items, dictionaries map keys to values, and functions package reusable steps.

Study part	Your notebook answer
Definition	Collections and functions means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Files and errors

Simple explanation: Read and write practice files, handle exceptions, and never collect or transmit real passwords or private data in beginner projects.

Study part	Your notebook answer
Definition	Files and errors means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 5 — Security Fundamentals

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: CIA triad

Simple explanation: Confidentiality limits access; integrity protects accuracy; availability keeps systems usable.

Study part	Your notebook answer
Definition	CIA triad means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Threats and risk

Simple explanation: A threat could cause harm, a vulnerability is a weakness, and risk combines likelihood and impact.

Study part	Your notebook answer
Definition	Threats and risk means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Authentication and access

Simple explanation: Authentication checks who you are; authorization checks what you may do. Use MFA and least privilege.

Study part	Your notebook answer
Definition	Authentication and access means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Security controls

Simple explanation: Preventive, detective, and corrective controls reduce the chance or impact of incidents.

Study part	Your notebook answer
Definition	Security controls means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 6 — Defensive Security & Logs

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: SOC overview

Simple explanation: A Security Operations Center monitors alerts, investigates suspicious activity, and coordinates response.

Study part	Your notebook answer
Definition	SOC overview means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Logs

Simple explanation: Logs record events. Learn timestamps, usernames, source IPs, event types, and context using synthetic sample data.

Study part	Your notebook answer
Definition	Logs means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Incident response

Simple explanation: A common cycle is preparation, detection/analysis, containment, eradication/recovery, and lessons learned.

Study part	Your notebook answer
Definition	Incident response means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: SIEM concepts

Simple explanation: A SIEM collects and correlates logs. Learn alert triage and false-positive handling using training data.

Study part	Your notebook answer
Definition	SIEM concepts means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 7 — Web & Application Security

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: How websites work

Simple explanation: Browsers and servers exchange HTTP requests and responses. HTML structures pages; CSS styles them; JavaScript adds behaviour.

Study part	Your notebook answer
Definition	How websites work means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: OWASP awareness

Simple explanation: Study common web risks conceptually: access-control failures, injection, insecure design, misconfiguration, and authentication weaknesses.

Study part	Your notebook answer
Definition	OWASP awareness means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Secure coding

Simple explanation: Validate inputs, encode output, use parameterized queries, protect secrets, and keep dependencies updated.

Study part	Your notebook answer
Definition	Secure coding means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Safe practice

Simple explanation: Use intentionally vulnerable training apps in an isolated lab, not real websites or other people's accounts.

Study part	Your notebook answer
Definition	Safe practice means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 8 — Cloud Fundamentals

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Cloud concepts

Simple explanation: Cloud services provide computing, storage, databases, and networking on demand.

Study part	Your notebook answer
Definition	Cloud concepts means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Shared responsibility

Simple explanation: The provider secures parts of the cloud platform; customers still configure identities, data, network access, and workloads correctly.

Study part	Your notebook answer
Definition	Shared responsibility means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Cloud identity

Simple explanation: Use least privilege, MFA, role-based access, short-lived credentials, and avoid public exposure of sensitive storage.

Study part	Your notebook answer
Definition	Cloud identity means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Cloud monitoring

Simple explanation: Review audit logs, alerts, backups, configuration checks, and budget limits.

Study part	Your notebook answer
Definition	Cloud monitoring means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 9 — Network Analysis & System Hardening

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Packet analysis

Simple explanation: A packet capture can help explain protocols and connections. Inspect only your own traffic or approved training captures.

Study part	Your notebook answer
Definition	Packet analysis means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Firewall concepts

Simple explanation: Firewalls allow or block traffic according to rules. Keep rules narrow and document why each rule exists.

Study part	Your notebook answer
Definition	Firewall concepts means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Hardening

Simple explanation: Remove unused services, patch software, disable unnecessary accounts, back up data, and review permissions.

Study part	Your notebook answer
Definition	Hardening means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Vulnerability management

Simple explanation: Identify assets, assess findings, prioritise by severity and context, remediate, then verify.

Study part	Your notebook answer
Definition	Vulnerability management means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 10 — Containers, DevSecOps & Automation

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Containers

Simple explanation: Containers package an application and its dependencies. Learn images, containers, registries, and isolation concepts.

Study part	Your notebook answer
Definition	Containers means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Secure development

Simple explanation: Shift security checks earlier: code review, dependency checks, secret scanning, and tests.

Study part	Your notebook answer
Definition	Secure development means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Automation

Simple explanation: Python can parse synthetic logs, format reports, and automate repetitive defensive checks.

Study part	Your notebook answer
Definition	Automation means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Supply chain

Simple explanation: Use trusted sources, pin and update dependencies, verify downloads, and protect build credentials.

Study part	Your notebook answer
Definition	Supply chain means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 11 — Portfolio & Capstone

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Project 1: log analyser

Simple explanation: Use a made-up log file; count failed logins by user/IP and generate a simple report. Do not monitor people without permission.

Study part	Your notebook answer
Definition	Project 1: log analyser means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Project 2: security checklist

Simple explanation: Create a checklist for laptop updates, MFA, backups, firewall, and account permissions.

Study part	Your notebook answer
Definition	Project 2: security checklist means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Project 3: phishing awareness

Simple explanation: Build a safe educational checker that explains warning signs from example URLs without visiting or attacking them.

Study part	Your notebook answer
Definition	Project 3: phishing awareness means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Documentation

Simple explanation: Include purpose, scope, tools, test data, screenshots only if available, results, limitations, and ethical boundaries.

Study part	Your notebook answer
Definition	Documentation means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Month 12 — Job Preparation & Review

Goal: understand the idea, explain it in your own words, and complete the practice task safely.

Lesson 1: Entry-level roles

Simple explanation: Explore IT support, service desk, network support, security internship, SOC analyst L1, and junior security analyst roles.

Study part	Your notebook answer
Definition	Entry-level roles means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 2: Resume and GitHub

Simple explanation: Show honest skills and projects, readable documentation, and what you personally built. Never claim certifications you do not have.

Study part	Your notebook answer
Definition	Resume and GitHub means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 3: Interview practice

Simple explanation: Explain DNS, IP, TCP/UDP, CIA triad, MFA, logs, phishing warning signs, and one project in simple words.

Study part	Your notebook answer
Definition	Interview practice means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Lesson 4: Final review

Simple explanation: Revise weak topics, complete capstone, practise labs legally, and make a realistic ongoing learning plan.

Study part	Your notebook answer
Definition	Final review means: _____
Example	A simple example is: _____
Why it matters	This matters because: _____
Common mistake	A mistake to avoid: _____

Practice: Use your own laptop or example data to draw a diagram/table or try a small exercise. Never test systems without permission.

Revision questions: (1) Explain this without reading. (2) Give one example. (3) What safe practice can you do today?

Quick Reference Comparison Tables

Hardware vs software

Concept	Meaning	Example
Hardware	Physical parts	Keyboard, RAM, SSD
Software	Programs/instructions	Windows, browser, Python

Authentication vs authorization

Concept	Meaning	Example
Authentication	Who are you?	Password, passkey, MFA
Authorization	What may you do?	Read-only access, admin permission

Threat vs vulnerability vs risk

Concept	Meaning	Example
Threat	Something that could cause harm	Phishing attempt
Vulnerability	A weakness	Outdated software
Risk	Likelihood and impact	Chance and consequences of misuse

TCP vs UDP

Concept	Meaning	Example
TCP	Reliable, ordered delivery	Many web/file-transfer uses
UDP	Lower overhead; no built-in delivery guarantee	Some live media and DNS use cases

Security controls

Type	Purpose	Example
Preventive	Try to stop incidents	MFA, secure configuration
Detective	Notice suspicious events	Alerts, monitoring
Corrective	Recover/reduce damage	Patching, restoring backups

CIA triad

Concept	Meaning	Example
Confidentiality	Only authorised access	Permissions, encryption
Integrity	Keep information accurate	Change controls
Availability	Keep systems usable	Backups, redundancy